# 1 The Six Trigonometric Functions

**Central question.** How do six functions arise naturally from the same moving point on a circle?

The six functions are easier to remember when we begin with one moving point rather than six separate formulas. Picture a point going around the unit circle. Its horizontal coordinate is $x = \cos\theta$ and its vertical coordinate is $y = \sin\theta$. The remaining functions compare those coordinates or take their reciprocals. Thus a single geometric position contains all six pieces of information.

**Learning goals.** By the end of this chapter, you should be able to define all six functions on the unit circle, find exact values in any quadrant, explain where each function is undefined, sketch their traditional graphs, and use trigonometry to interpret measured lengths and angles.

## 1.1 From a point to six functions

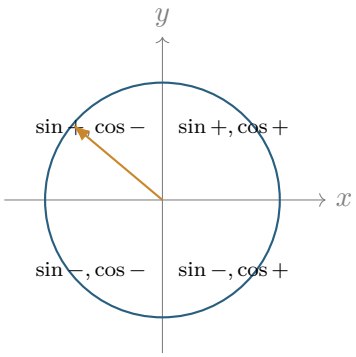
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

These are not six unrelated measurements. They are six different ways of reading the same geometric information.

**Definition 1.1.** The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

**Example 1.1.** At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

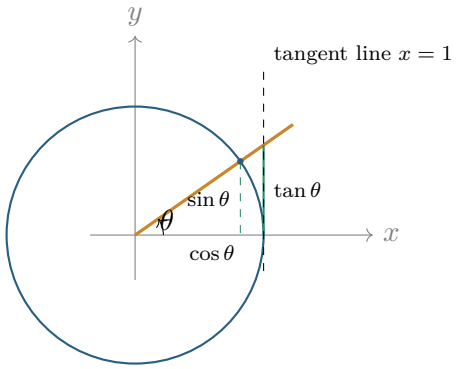



## 1.2 Why tangent is a slope

Imagine the ray from the origin through the point $P = (\cos\theta, \sin\theta)$. Where cosine is nonzero, the slope of that ray is

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

This connects a familiar idea from linear equations to circular geometry: tangent records the *directional slope* of the ray. A nearly vertical ray has a very large positive or negative slope. A vertical ray has no finite slope at all.




On the vertical line $x = 1$, the extension of the ray reaches height $\tan\theta$. This construction shows why the function is named *tangent*: it can be read along a line tangent to the unit circle.

**Example 1.2.** At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

The same slope describes a ramp rising one unit for every $\sqrt{3}$ units of horizontal travel.

## 1.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

**Example 1.3.** For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

Notice that no signs were memorized separately. All of them followed from the two coordinates.

| | sin | cos | tan | csc | sec | cot |
|---|---|---|---|---|---|---|
| Quadrant I | + | + | + | + | + | + |
| Quadrant II | + | − | − | + | − | − |
| Quadrant III | − | − | + | − | − | + |
| Quadrant IV | − | + | − | − | + | − |

**Try it.** The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

### 1.4 Where are the functions undefined?

Division by zero has no meaning. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer $k$. The word *undefined* matters: an undefined value is not the number zero.

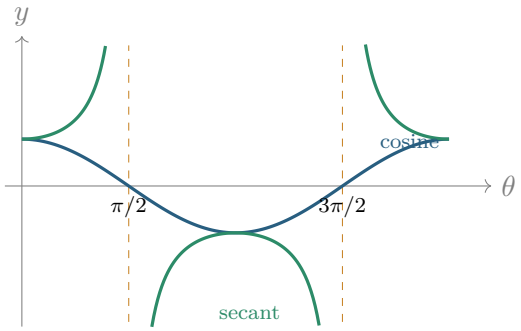
**Try it.** Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

### 1.5 Zeros, poles, and disconnected graphs

A denominator becomes small as its reciprocal becomes large. This is the main visual idea behind tangent, cotangent, secant, and cosecant.

For example, as $\theta$ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

On an ordinary graph, the vertical line $x = \pi/2$ is a *vertical asymptote* of secant and tangent. The picture shows why a graph cannot be joined through that line.




Here the green curve has been drawn only where its values fit in the viewing window. It continues upward or downward without touching its asymptotes. The graph does *not* cross the asymptote, and we must not connect the separated branches.

**Try it.** Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

### 1.6 Right triangles are a special case

For an acute angle $\theta$ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse $h$ gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit circle is more general than the right-triangle definitions: it handles negative angles, obtuse angles, and angles exceeding a full revolution.

**Example 1.4.** A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

This gives us the complete right-triangle set of six ratios:

| | opposite and hypotenuse | adjacent and hypotenuse | opposite and adjacent |
|---|---|---|---|
| direct | $\sin\theta = o/h$ | $\cos\theta = a/h$ | $\tan\theta = o/a$ |
| reciprocal | $\csc\theta = h/o$ | $\sec\theta = h/a$ | $\cot\theta = a/o$ |

All six ratios come from the same three lengths. The unit-circle definitions extend them beyond acute angles without changing the relationships.

**Example 1.5.** A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If $\theta$ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

The reciprocal values describe the same geometry, not new triangles.

## 1.7 Special angles without memorizing a disconnected table

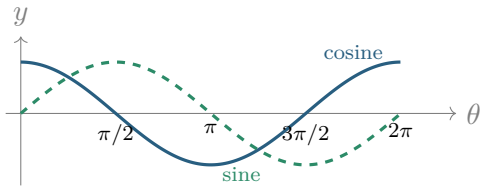
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.




## 1.8 Reading the traditional trigonometric graphs

A graph changes the question we ask. On the unit circle, we hold an angle fixed and locate a point. On a Cartesian graph, we let the angle vary along the horizontal axis and watch one numerical output rise and fall. The graph of a trigonometric function is not the circular path of the point; it is a record of one coordinate or ratio.

Sine and cosine remain between $-1$ and 1. Both repeat after a full rotation, so their fundamental period is $2\pi$ radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is $\pi$ radians, or 180°. Secant and cosecant share the $2\pi$ period of their reciprocal partners.




**Three questions to ask of any trig graph.** Where does the function cross the horizontal axis? Where is it undefined? How far must the input angle increase before the graph repeats? Together, these questions reveal most of its global shape.

**Example 1.6.** Why does tangent repeat after only $\pi$? Rotating the unit-circle point by 180° changes both coordinate signs:

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

The quotient repeats even though the point moves to the opposite side of the circle.

**Explore interactively.** The course's Six Trigonometric Graphs Explorer lets you compare all six functions. The Reciprocal Trigonometric Graphs Explorer places each function beside its reciprocal; watch how the zeros of one function become asymptotes of the other.

**Try it.** Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

## 1.9 Applications: orientation and measurement

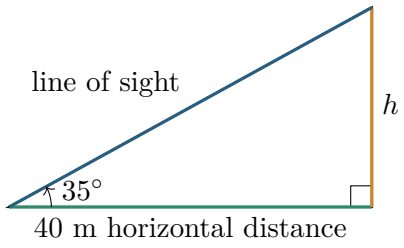
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is $35^\circ$. The tower's height $h$ satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

## 1.10 Measuring height and choosing the right ratio

In a physical measurement problem, the right triangle should come *before* the calculator. A horizontal distance and a vertical height are related by tangent. A direct line-of-sight distance and a vertical height are related by sine. Recognizing the geometry is more important than memorizing which button to press.




**Worked example: estimate a tree height.** An observer is 20 meters from the base of a tree and measures an angle of elevation of $40^\circ$ to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The final addition matters: a correct trig ratio still needs an accurate interpretation of the measured height.

**Try it.** A rescue drone is observed at an elevation angle of $25^\circ$ from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

## 1.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.

4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?

5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.

6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.

7. A drone is viewed at an elevation angle of $28^\circ$ from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.

9. Find the exact values of all six trig functions at $11\pi/6$.

10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.

11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.

12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.

13. If $\cos\theta = -12/13$ and $\theta$ lies in quadrant II, determine all other trig values exactly.

14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.

15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.